

**AGA KHAN UNIVERSITY EXAMINATION BOARD
HIGHER SECONDARY SCHOOL CERTIFICATE**

CLASS XI

ANNUAL EXAMINATIONS 2022

Physics

Total Time: 2 hours 10 minutes

Total Marks: 65 (50-Theory & 15-Alternate to Practical)

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.

4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.

Correct Way	Incorrect Ways
1 ☐ A ☐ B ☒ C ☐ D	1 ☐ A ☐ B ☒ C ☐ D
	2 ☐ A ☐ B ☒ C ☐ D
	3 ☐ A ☐ B ☒ C ☐ D
	4 ☐ A ☐ B ☒ C ☐ D

Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

THEORY (Questions 1-50)

1. The type of error, which is produced by a faulty measuring instrument is called

A. logical error.
 B. random error.
 C. personal error.
 D. systematic error.

2. Consider the basic dimensional formula as $M^x L^y T^z$.

Which of the following values of x , y and z will satisfy the physical quantity of torque?

	x	y	z
A	2	-1	2
B	1	2	-2
C	2	1	2
D	1	-2	-2

3. The multiplication product of 20.21 with 20.22 upto four significant figures is

A. 408.6
 B. 408.64
 C. 408.6462
 D. 409.0

4. In a champions' trophy of dart match, a player was given three chances to hit the centre target with darts. He managed to hit all the darts close to each other but away from the centre.

The given case depicts that the player's shots were

A. precise.
 B. accurate.
 C. uncertain.
 D. unreliable.

5. If a bridge is anchored on two concrete pillars, then the net torque on the bridge will be



- A. $\tau = 0$
- B. $0 < \tau < 1$
- C. $\tau = 1$
- D. $1 < \tau < 2$

6. If an object satisfies the first condition of equilibrium, then the rightward and leftward forces will be

	Magnitude	Direction
A	equal	opposite
B	unequal	same
C	equal	same
D	unequal	opposite

7. In a Cartesian Coordinate System, the points $(-6, -8)$ lies in the

- A. I quadrant.
- B. II quadrant.
- C. III quadrant.
- D. IV quadrant.

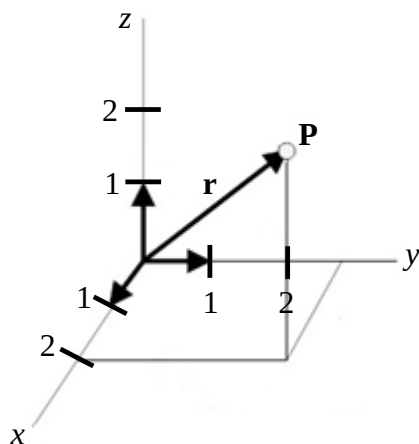
8. The mathematical operation of cross product is performed between

- A. two scalar quantities.
- B. two vector quantities.
- C. one scalar and one numeric value.
- D. one scalar and one vector quantity.

9. The dot product of two perpendicular unit vectors is equal to

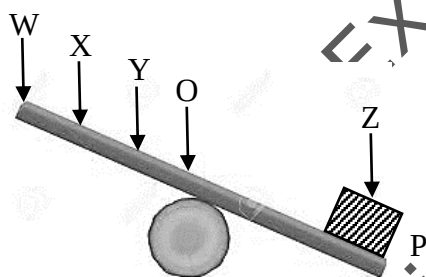
- A. -1
- B. 0
- C. 1
- D. $\sqrt{2}$

10. In the given vector diagram, the point **P** is represented by the vector **r**.



The vector **r** represents a/ an

- A. unit vector.
 B. null vector.
 C. equal vector.
 D. position vector.
11. A wooden slab is balanced on a circular log of wood at point (O) in the given diagram. Object (P) is placed at point (Z).



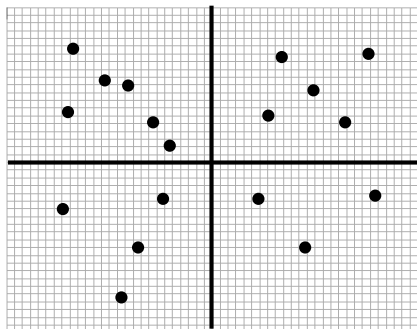
Zohan is given the task to throw the object (P) high in the air by

- I. jumping at any position (W, X, Y) on the slab
 II. moving or not moving the position of log at point (O)

Which of the following options should Zohan pick to complete the task?

	Jumping Position	Wooden Log Position
A	W	O
B	W	X
C	X	Y
D	Y	O

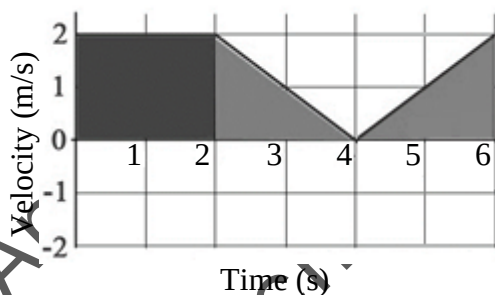
12. The highest number of points lies in the



- A. I quadrant.
B. II quadrant.
C. III quadrant.
D. IV quadrant.
13. Which of the following laws of conservation is used in the propulsion of rockets?
- A. Heat
B. Mass
C. Energy
D. Momentum

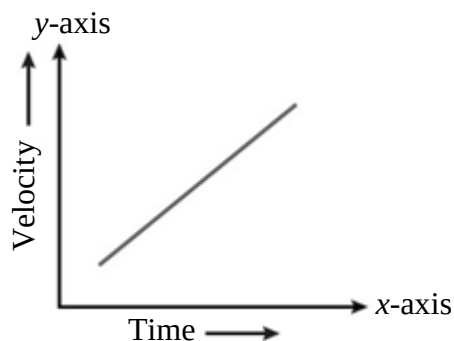
Use the given graph to answer Q.14 and Q.15.

The given figure represents a velocity-time graph of a moving car.

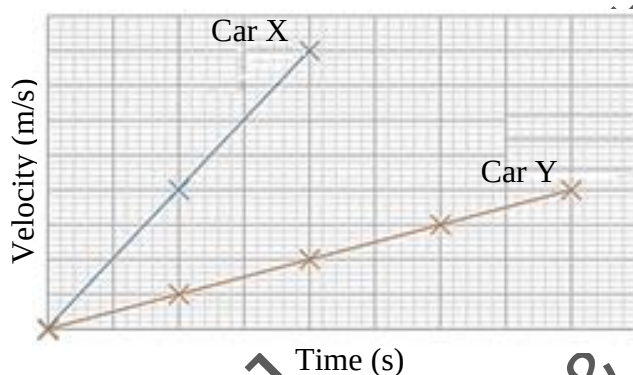


14. The maximum velocity by the car during its journey is
- A. -1 m/s .
B. 2 m/s .
C. 4 m/s .
D. 6 m/s .
15. The difference in the displacement from the starting point to 2 s and from 2 s to 5 s is
- A. 1 m.
B. 1.5 m.
C. 2.5 m.
D. 5 m.

16. The type of motion represented by the given velocity-time graph is



- A. variable motion.
 B. instantaneous motion.
 C. uniformly accelerated motion.
 D. non-uniformly accelerated motion.
17. The given velocity-time graph shows two different cars that accelerate on a long straight road.



Which of the following statement(s) is/ are CORRECT about the given scenario?

- I. Car Y has covered more displacement
 II. Both cars have started their journey from rest
 III. Car X has taken more time to finish its journey
- A. I only.
 B. II only.
 C. I and III.
 D. II and III.
18. A freely falling ball bounces back when it hits the ground. At the point where the ball hits the ground, the impulse on the ball as compared to the momentum of the ball will be
- A. less.
 B. zero.
 C. equals.
 D. greater.

19. If the different layers of a fluid offer no resistance over each other, then the fluid will be a/ an
- A. ideal fluid.
 - B. viscous fluid.
 - C. Newtonian fluid.
 - D. non-viscous fluid.
20. In fluid dynamics, if every particle passing through a given point undergoes in an irregular pattern and fluctuation, then this type of flow is called a
- A. steady flow.
 - B. laminar flow.
 - C. turbulent flow.
 - D. streamline flow.
21. In a fire fighting machine, a jet is attached to the head of water pipes to increase the
- A. mass of water flowing out of it.
 - B. volume of water flowing out of it.
 - C. velocity of water flowing out of it.
 - D. distance covered by water flowing out of it.
22. When an object moves with a terminal velocity, then the acceleration of the object will
- A. -9.8 m/s^2 .
 - B. 0 m/s^2 .
 - C. 4.9 m/s^2 .
 - D. 9.8 m/s^2 .
23. An industrial liquid flows through a section of a pipe with an area of cross-section 2 m^2 at a velocity of 10 m/s . If at a certain section the area of cross-section decreases to 1 m^2 , then the velocity of the liquid will be
- A. 5 m/s .
 - B. 10 m/s .
 - C. 15 m/s .
 - D. 20 m/s .
24. According to the Bernoulli's principle, when the pressure decreases during the flow of fluid through different sections of a pipe, then the speed of the fluid will
- A. increase.
 - B. decrease.
 - C. become zero.
 - D. remain the same.

25. The time taken by a simple pendulum to complete one oscillation can be increased, if the

- I. value of 'g' is increased
- II. value of 'g' is decreased
- III. length of the pendulum is increased
- IV. length of the pendulum is decreased

- A. I and III.
- B. I and IV.
- C. II and III.
- D. II and IV.

26. When the length of a simple pendulum increases, its frequency will

- A. increase.
- B. decrease.
- C. become zero.
- D. remain constant.

27. All of the following are considered as everyday examples of resonance EXCEPT

- A. a playground swing.
- B. shock absorbers of a car.
- C. turning the knob of the radio set.
- D. a music system playing on a heavy beat.

28. When the bob of a simple pendulum is at the extreme position and the mean position, then its kinetic energy will be

	Extreme Position	Mean Position
A	maximum K.E.	minimum K.E.
B	maximum K.E.	maximum K.E.
C	minimum K.E.	minimum K.E.
D	minimum K.E.	maximum K.E.

29. A simple pendulum having time period **T** is suspended from the ceiling of an elevator that is at rest position.

When the elevator falls freely, then the time period of the pendulum will be

- A. $\frac{T}{2}$.
- B. T.
- C. 2T.
- D. infinite.

30. If the length of a simple pendulum is 0.4 m and the mass of pendulum's bob is 1 kg, then the time period of the pendulum will be

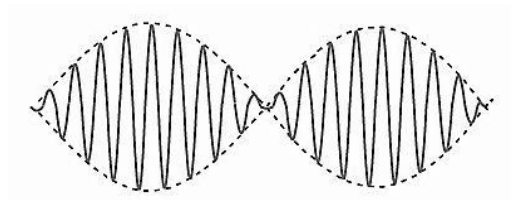
(**Note:** Take the acceleration due to gravity as 9.8 m/s^2 .)

- A. 0.405 s.
 - B. 1.269 s.
 - C. 2.007 s.
 - D. 12.44 s.
31. In wave motion, the distance between a consecutive node and an antinode is
- A. $\frac{\lambda}{4}$.
 - B. $\frac{\lambda}{2}$.
 - C. λ .
 - D. 2λ .
32. A wave with a wavelength of 2 m having frequency 0.5 Hz will travel with the velocity of
- A. 1 m/s.
 - B. 1.5 m/s.
 - C. 2.5 m/s.
 - D. 4 m/s.
33. Doppler's effect can also be used to calculate the speed of
- A. light.
 - B. water waves.
 - C. expanding galaxies.
 - D. electromagnetic waves.
34. One of the important phenomena of the superposition of sound waves, beats, is an example of
- A. refraction.
 - B. diffraction.
 - C. polarisation.
 - D. interference.
35. In summer season, with the rise of temperature, the velocity of sound
- A. increases.
 - B. decreases.
 - C. becomes zero.
 - D. remains constant.

36. The presence of stationary waves on a string can easily be detected at

- I. nodes
- II. antinodes
- A. I only.
- B. II only.
- C. I and II.
- D. between I and II.

37. The given diagram shows a phenomena of wave production known as



- A. beats.
 - B. nodes.
 - C. destructive interference.
 - D. constructive interference.
38. Doppler's effect is used in sonar and artificial satellites. Therefore, it can be applied to
- I. light waves
 - II. sound waves
 - III. electromagnetic waves
 - A. I only.
 - B. III only.
 - C. I and II.
 - D. II and III.
39. If an aeroplane approaches a radar system, then the wavelength of the communication wave reflected from the aeroplane to the ground station would
- A. be longer.
 - B. be shorter.
 - C. become infinite.
 - D. remain the same.
40. According to Boyle's law, if the pressure is doubled, then the volume of a gas becomes
- A. one-fourth.
 - B. one-halved.
 - C. two times.
 - D. three times.

41. The amount of heat energy needed to raise the temperature of 1 kg of a substance through 1 K is called
- A. specific heat.
 - B. heat capacity.
 - C. heat of fusion.
 - D. heat of vapourisation.
42. Which of the following is the CORRECT sequence of the working principle of Carnot engine?
- A. Adiabatic compression → isothermal expansion → isobaric compression → isochoric expansion
 - B. Isothermal expansion → isobaric compression → adiabatic compression → isochoric expansion
 - C. Isothermal expansion → adiabatic expansion → isothermal compression → adiabatic compression
 - D. Isobaric compression → isothermal expansion → isochoric expansion → adiabatic compression
43. All of the following are the examples of reverse heat engine EXCEPT
- A. heat pump.
 - B. refrigerator.
 - C. electric boiler.
 - D. air conditioner.
44. For an ideal gas at a constant temperature, if a graph between variable values of volume (V) and pressure $\left(\frac{1}{P}\right)$ is plotted, then the shape of the graph will be
- A. ellipse.
 - B. parabola.
 - C. hyperbola.
 - D. straight line.
45. The process in which the volume of a system remains constant is called
- A. isobaric process.
 - B. adiabatic process.
 - C. isochoric process.
 - D. isothermal process.
46. The efficiency of a heat engine that operates between temperatures of 1500 K in the fire chamber and 600 K in the exhaust chamber is
- A. 25 %.
 - B. 50 %.
 - C. 60 %.
 - D. 65 %.

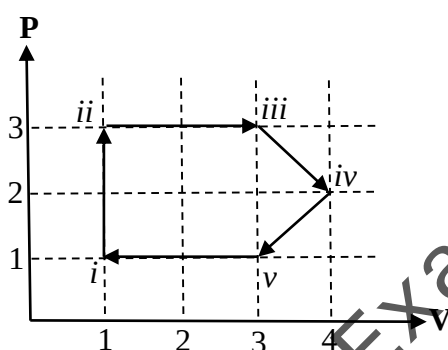
47. An ideal gas confined in a container initially has pressure **P**. Suppose, the absolute temperature of the gas becomes two times and the volume of the container becomes four times.

The pressure will then be

- A. $\frac{P}{4}$.
- B. $\frac{P}{2}$.
- C. P .
- D. $2P$.

Use the given graph to answer Q.48 and Q.49.

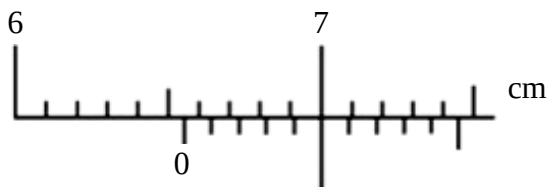
An ideal gas undergoes a cyclic process as shown in the given pressure **P** versus volume **V** graph.



48. During which part of the process no work is done on or by the gas?
- A. *i* to *ii*
 - B. *ii* to *iii*
 - C. *iii* to *iv*
 - D. *iv* to *v*
49. The gas is at its highest temperature at point(s)
- A. *ii* only.
 - B. *iv* only.
 - C. *i* and *ii*.
 - D. *iii* and *v*.
50. An ideal gas is made up of **x** number of diatomic molecules and each molecule has mass **M**.
- All of the following statements about this gas are TRUE EXCEPT the
- A. molecules have the same velocity.
 - B. molecules make elastic collisions with each other.
 - C. molecules make elastic collisions with the walls of the container.
 - D. temperature of the gas is proportional to the average kinetic energy.

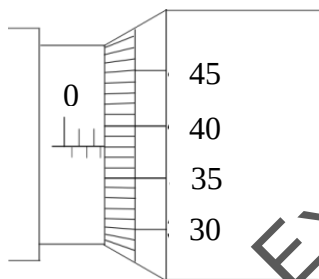
ALTERNATE TO PRACTICAL (ATP: Questions 51-65)

51. The given diagram shows two different scales on a Vernier callipers.



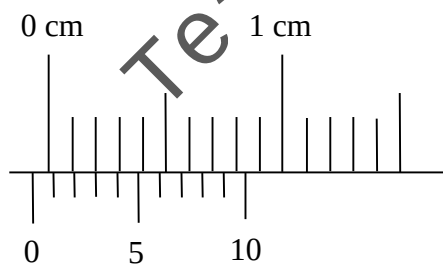
The reading on the main scale is

- A. 6 cm.
 B. 6.5 cm.
 C. 7 cm.
 D. 7.4 cm.
52. The given figure shows two different scales on a screw gauge.



The circular scale reading is

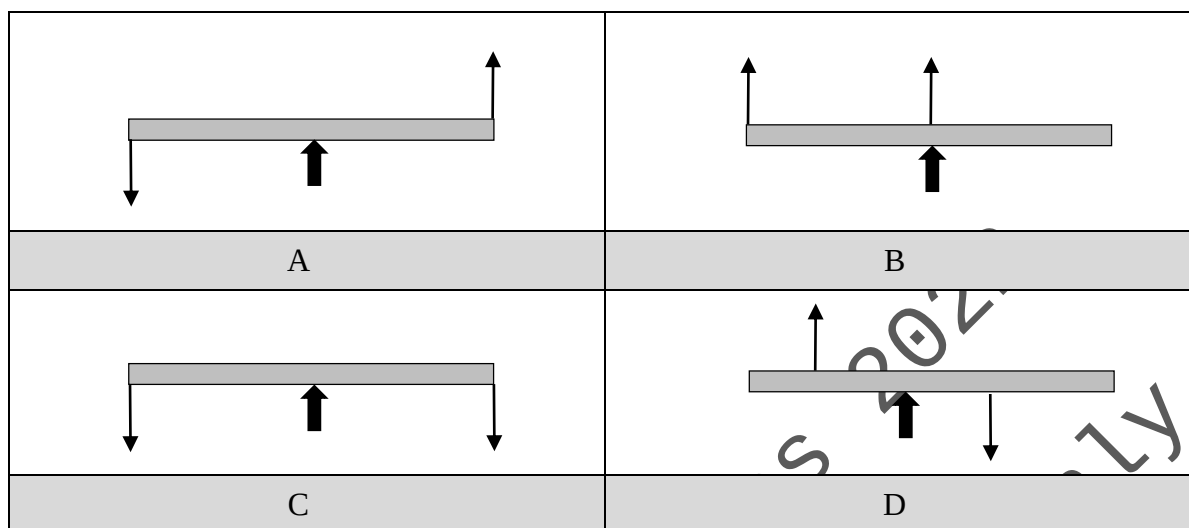
- A. 35
 B. 38
 C. 40
 D. 45
53. The zero error in the given measurement is



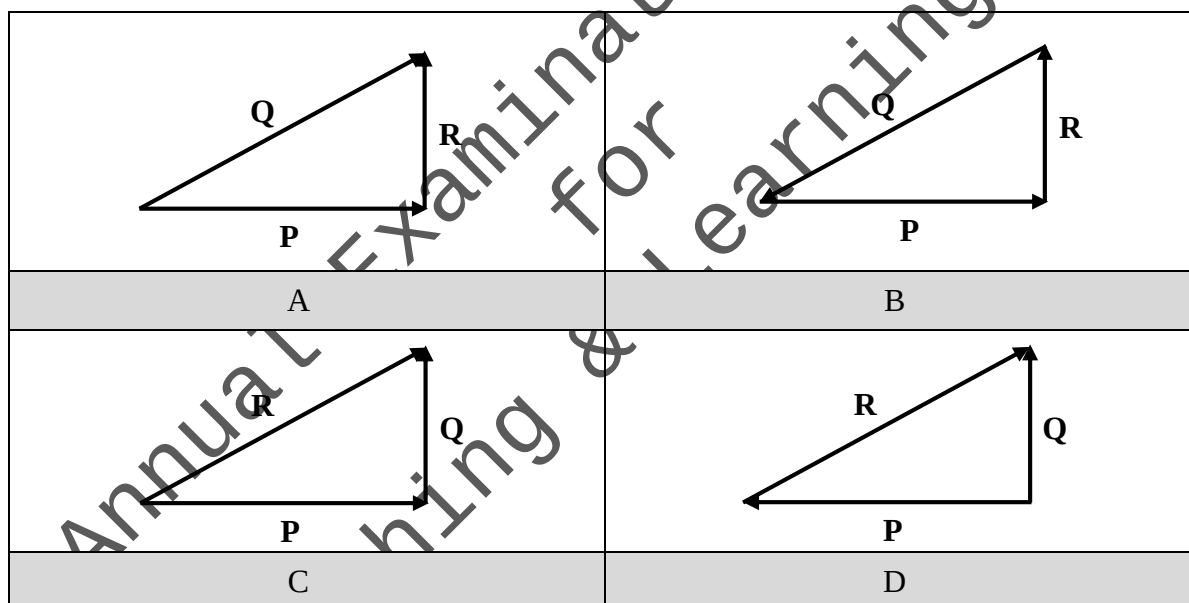
- A. - 0.3 mm.
 B. - 0.03 mm.
 C. 0.03 mm.
 D. 0.3 mm.

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54. A uniform rod is fixed at its midpoint on a fulcrum. When two equal forces are applied on the rod then the diagram that represents the equilibrium position is



55. By adding two vectors **P** and **Q**, the CORRECT representation of the resultant **R** is



56. The height of a mobile phone tower is 125 m. If a device is dropped from the tower, it reaches the ground in 5 s. The value of acceleration of the device just before hitting the ground
- will be 10 m/s^2 .
 - will be 25 m/s^2 .
 - will be 50 m/s^2 .
 - cannot be determined.

57. Which of the following pieces of ticker tape shows that the object is speeding up its motion?

(**Note:** Suppose that the left hand side of the ticker tape shows the initial point of the tape.)

- A. 
- B. 
- C. 
- D. 

58. The time period of a simple pendulum is directly and inversely proportional to the square root of its

	Directly Proportional	Inversely Proportional
A	mass	amplitude
B	amplitude	length
C	mass	acceleration due to gravity
D	length	acceleration due to gravity

59. If a student plans to make a simple pendulum of time period 2 s, then the length of the pendulum will be

(**Note:** Take the value of acceleration due to gravity as 9.8 m/s^2 .)

- A. 0.996 m.
 B. 1.282 m.
 C. 1.992 m.
 D. 3.120 m.
60. The beat frequency of two waves produced by two different tuning forks 505 Hz and 512 Hz will be
- A. 7 Hz.
 B. 6 Hz.
 C. 4 Hz.
 D. 0 Hz.
61. In an experiment to investigate the laws of vibration of stretched strings by sonometre, all of the following factors affect the frequency of a tuning fork EXCEPT the
- A. length of the prong.
 B. force on the tuning fork.
 C. mass attached to the prong.
 D. nature of the material of the tuning fork.

62. The distance between two consecutive antinodes of an incident wave is 1 m.

If the frequency of the wave is 25 Hz, then the velocity of the wave will be

- A. 12.5 m/s.
- B. 25 m/s.
- C. 50 m/s.
- D. 100 m/s.

63. In an experiment of measuring the mechanical equivalent of heat, a stirrer is placed in the calorimetre filled with hot water.

After some time, the exposed end of the stirrer becomes hot.

The process(es) that is/ are mainly involved in the transferring of heat processes is/ are

- I. radiation
- II. convection
- III. conduction

- A. I only.
- B. III only.
- C. I and II.
- D. II and III.

64. A student touches the metal surface of a hot calorimetre. He is likely to burn his hand.

If he holds his hand in the hot air inside the calorimetre, then he will remain fine.

Which of the following statements BEST describe the given condition?

- A. Both the metal and air are poor conductor of heat.
- B. Both the metal and air are good conductors of heat.
- C. Metal is a good conductor, but air is a poor conductor of heat.
- D. Air is a good conductor, but metal is a poor conductor of heat.

65. In a practical examination a student has been given a task to find the specific heat of 0.5 kg of water placed at a burning burner where thermal energy of 1000 J is used to raise its temperature from 80°C to 90°C.

The specific heat of water will be

- A. 200 J/kg°C.
- B. 100 J/kg°C.
- C. 22.2 J/kg°C.
- D. 11.7 J/kg°C.

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